Proposition 10: a factor 2 gap for the cut distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition10_CutLowerBound` formalises Proposition 10 (p. 9) with the construction of Appendix E.1, Lemmas 33 and 34, the remark after Proposition 10 ($d_{\square}^{\mathrm{OT}}(G,H) = 4$; its Lean certificate, linked in footnote 2 of the paper, is `OptimalAlignment.lean`) and Theorem 3 (3).

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Prop10.proposition10`: the graphs G, H of Appendix E.1 have 145 vertices,

$$d_{\square}(G,H) = 8 \qquad \text{and} \qquad d_{\square}^{\mathrm{OT}}(G,H) \le 4,$$

exactly as Proposition 10 states. `GsimOT.Prop10.dOTCut_eq_four`: $d_{\square}^{\mathrm{OT}}(G,H) = 4$, the Lean certificate of the remark after Proposition 10. `GsimOT.Prop10.theorem3_item3`: $\delta_{\square}(G,H) = 2\,\delta_{\square}^{\mathrm{OT}}(G,H)$ (Theorem 3 (3)).

Correspondence

Paper	Page	Statement	Lean declaration(s) and location

Proposition 10	9	Graphs on 145 vertices with $d_{\square}(G,H) = 8$ and $d_{\square}^{\mathrm{OT}}(G,H) \le 4$ (exactly as stated)	`GsimOT.Prop10.proposition10`
			`Proposition10_CutLowerBound/Proposition10.lean:20`
			`GsimOT.Prop10.card_Vtx`
			`Proposition10_CutLowerBound/Construction.lean:40`

Remark after Prop. 10	10	$d_{\square}^{\mathrm{OT}}(G,H) = 4$ (l. 330; the Lean certificate `OptimalAlignment.lean`, linked in footnote 2); $C_{\square}(Q) \ge 4$ for every $Q \in \mathrm{DS}(V,W)$	`GsimOT.Prop10.dOTCut_eq_four`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:412`
			`GsimOT.Prop10.four_le_dOTCut`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:406`
			`GsimOT.Prop10.four_le_costCut`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:371`

Theorem 3 (3)	4	$\delta_{\square}(G,H) = 2\,\delta_{\square}^{\mathrm{OT}}(G,H)$ for these graphs (l. 126)	`GsimOT.Prop10.theorem3_item3` `Proposition10_CutLowerBound/Proposition10.lean:26`

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Appendix E.1	30–31	Core pairs v_i^σ, markers L_i ($\lvert L_i\rvert = 9(i+1)$), matrix Z	`GsimOT.Prop10.Vtx`
			`Proposition10_CutLowerBound/Construction.lean:38`
			`GsimOT.Prop10.Core`
			`Proposition10_CutLowerBound/Construction.lean:32`
			`GsimOT.Prop10.Marker`
			`Proposition10_CutLowerBound/Construction.lean:35`
			`GsimOT.Prop10.Z`
			`Proposition10_CutLowerBound/Construction.lean:50`

Appendix E.1, Fig. 3	10, 31	G: $v_i^\sigma v_j^\tau \in E$ iff $\sigma\tau = Z_{ij}$; H: iff $\sigma\tau = -Z_{ij}$; markers adjacent to their pair	`GsimOT.Prop10.gAdj`
			`Proposition10_CutLowerBound/Construction.lean:58`
			`GsimOT.Prop10.hAdj`
			`Proposition10_CutLowerBound/Construction.lean:65`
			`GsimOT.Prop10.G12`
			`Proposition10_CutLowerBound/Construction.lean:90`
			`GsimOT.Prop10.H12`
			`Proposition10_CutLowerBound/Construction.lean:96`

Appendix E.1	31	Degrees $13, 22, 31, 40, 49$ on pairs, 2 on markers; different degrees differ by ≥ 9	`GsimOT.Prop10.dg`
			`Proposition10_CutLowerBound/Construction.lean:170`
			`GsimOT.Prop10.sum_adj_G12`
			`Proposition10_CutLowerBound/Construction.lean:191`
			`GsimOT.Prop10.sum_adj_H12`
			`Proposition10_CutLowerBound/Construction.lean:202`
			`GsimOT.Prop10.nine_le_of_dg_ne`
			`Proposition10_CutLowerBound/Construction.lean:213`

Lemma 33	31	$d_\square(G, H) = 8$	`GsimOT.Prop10.lemma33` `Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:189`
Proof of Lemma 33	31	$C_\square(\pi) \geq \lvert \deg_G(v) - \deg_H(\pi(v))\rvert$ (cut $S = \{\pi v\}$, $T = W$)	`GsimOT.Prop10.abs_dg_sub_le` `Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:80`

Paper	Page	Statement	Lean declaration(s) and location

Proof of Lemma 33	31	Pair-preserving π: core cut values $x^\top My$ with $M_{ij} = \frac{1}{2}Z_{ij}(1 + r_i r_j)$; ≥ 8 for all 32 patterns r	`GsimOT.Prop10.Mr`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:88`
			`GsimOT.Prop10.wit`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:91`
			`GsimOT.Prop10.eight_le_wit`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:130`
			`GsimOT.Prop10.eight_le_of_dg`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:155`

Proof of Lemma 33	31	Every bijection costs ≥ 8	`GsimOT.Prop10.eight_le_cutNorm` `Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:181`

Proof of Lemma 33	31	For $\pi = \mathrm{id}$ every cut value is $x^\top Zy$, $x, y \in \{-1, 0, 1\}^5$, and $\lvert x^\top Zy\rvert \leq 8$	`GsimOT.Prop10.cut_sum_eq_core`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:25`
			`GsimOT.Prop10.abs_bilin_Z_le`
			`Proposition10_CutLowerBound/Construction.lean:264`
			`GsimOT.Prop10.cutNorm_refl_le`
			`Proposition10_CutLowerBound/Lemma33_BijectionCost.lean:66`

Lemma 34	31	$d_{\square}^{\mathrm{OT}}(G, H) \leq 4$	`GsimOT.Prop10.lemma34` `Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:173`

Proof of Lemma 34	31	Q: identity on markers, $\frac{1}{2}$ on each pair block	`GsimOT.Prop10.Q40`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:22`
			`GsimOT.Prop10.Q40_isDS`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:39`

Proof of Lemma 34	32	Discrepancies involving markers vanish; core discrepancy $\frac{1}{2}Z_{ij}(\sigma\sigma' + \tau\tau')$	`GsimOT.Prop10.B40_eq_zero_left`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:81`
			`GsimOT.Prop10.B40_eq_zero_right`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:92`
			`GsimOT.Prop10.B40_atom`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:104`

Proof of Lemma 34	32	Straight/crossed decomposition: two pentagon copies of weight $\frac{1}{2}$, each cut ≤ 2	`GsimOT.Prop10.atom_sum_eq`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:122`
			`GsimOT.Prop10.costCut_Q40_le`
			`Proposition10_CutLowerBound/Lemma34_FractionalCost.lean:140`

Paper	Page	Statement	Lean declaration(s) and location

$d_{\square}^{\mathrm{OT}} \geq 4$, step 1		$\lvert B(f,g)\rvert \leq C_{\square}(Q)$ for $f,g\colon \Omega \to [0,1]$	`GsimOT.Prop10.exists_abs_sum_mul_le`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:46`
			`GsimOT.Prop10.abs_bilin_le_cutNorm`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:63`

$d_{\square}^{\mathrm{OT}} \geq 4$, step 2		$\sum_y D(x,y)q_y = d(x)$ and $\sum_x q_x d(x) = 0$	`GsimOT.Prop10.sum_disc_mul`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:94`
			`GsimOT.Prop10.sum_B_right`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:106`
			`GsimOT.Prop10.sum_mul_dd`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:111`

$d_{\square}^{\mathrm{OT}} \geq 4$, step 3		$L + B(s,t) \leq 4C_{\square}(Q)$ for $\lvert s\rvert, \lvert t\rvert \leq 1$, $s = 0$ where $d \neq 0$	`GsimOT.Prop10.degree_add_bil_le` `Proposition10_CutLowerBound/OptimalAlignment.lean:123`

$d_{\square}^{\mathrm{OT}} \geq 4$, step 4		$L \geq 9(10 - T)$, T the mass on the core atoms	`GsimOT.Prop10.nine_mul_le_row`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:324`
			`GsimOT.Prop10.nine_mul_le_L`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:348`

$d_{\square}^{\mathrm{OT}} \geq 4$, step 5		$\max_{a,b\in\{\pm 1\}^5} a^{\top}\mathrm{diag}(x)Z\mathrm{diag}(x)b \geq \frac{16}{5}\sum_i x_i - 8$ for $x \in [0,2]^5$	`GsimOT.Prop10.zForm`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:187`
			`GsimOT.Prop10.zForm_flip`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:191`
			`GsimOT.Prop10.exists_zForm_ge`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:200`

$d_{\square}^{\mathrm{OT}} \geq 4$, step 5		Test functions on straight and crossed atoms: $B(s,t) = a^{\top}CZCb + a'^{\top}HZHb'$	`GsimOT.Prop10.testFn`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:232`
			`GsimOT.Prop10.disc_atom`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:265`
			`GsimOT.Prop10.bil_testFn`
			`Proposition10_CutLowerBound/OptimalAlignment.lean:277`

Shared notions used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location

Appendix A	14	Cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$	`GsimOT.cutNorm`
			`Common/CutNorm.lean:32`
			`GsimOT.le_cutNorm`
			`Common/CutNorm.lean:39`
			`GsimOT.cutNorm_le`
			`Common/CutNorm.lean:44`
			`GsimOT.exists_eq_cutNorm`
			`Common/CutNorm.lean:52`
			`GsimOT.cutNorm_smul`
			`Common/CutNorm.lean:59`

Section 1, Section 2	2, 5	$B(G,H,Q)$ and $C_\square(Q) = \|B(G,H,Q)\|_\square$	`GsimOT.B`
			`Common/OTDistances.lean:58`
			`GsimOT.costCut`
			`Common/CutNorm.lean:79`

Section 1, Section 2	1–2, 5	$d_\square$, $d_\square^{\mathrm{OT}}$ (infimum) and $\delta_\square = n^{-2} d_\square$, $\delta_\square^{\mathrm{OT}} = n^{-2} d_\square^{\mathrm{OT}}$	`GsimOT.dCut`
			`Common/CutNorm.lean:82`
			`GsimOT.dOTCut`
			`Common/CutNorm.lean:85`
			`GsimOT.deltaCut`
			`Common/CutNorm.lean:88`
			`GsimOT.deltaOTCut`
			`Common/CutNorm.lean:91`

Appendix E.2, Eq. (21), E.2.2	32, 36	$\zeta_R(A,B) = \|D_{A,B}\|_{\square,\mu_R}$ with $\mu_R = R/n$;
		$\zeta_R(A_G, A_H) = \zeta_R = n^{-2} C_\square(R)$ for graphs
			`GsimOT.epsR`
			`Common/CutNorm.lean:105`
			`GsimOT.adjMat`
			`Common/CutNorm.lean:94`
			`GsimOT.epsR_adjMat`
			`Common/CutNorm.lean:110`

Paper	Page	Statement	Lean declaration(s) and location
Section 1, Appendix B	1, 3, 15	$\kappa_{p\to q} = 1/q + 1 - 1/p$, $A_G^{\pi} - A_H$, $d_{p\to q}$ and $\delta_{p\to q} = n^{-\kappa_{p\to q}} d_{p\to q}$ $(s(n) = n^{\kappa_{p\to q}})$	`GsimOT.kappa` `Common/AlignedDistances.lean:23` `GsimOT.alignedDiff` `Common/AlignedDistances.lean:31` `GsimOT.distPQ` `Common/AlignedDistances.lean:35` `GsimOT.deltaPQ` `Common/AlignedDistances.lean:39`

Notes on the formalisation

1. $d_{\square}^{\mathrm{OT}}(G,H) = 4$**.** Proposition 10 states $d_{\square}^{\mathrm{OT}}(G,H) \le 4$. The remark after it (p. 10, l. 330) states equality, verified by "one of our Lean certificates"; its footnote 2 links the file `OptimalAlignment.lean`, whose main result is `dOTCut_eq_four`. Theorem 3 (3) (p. 4, l. 126) uses this equality. The paper refers to this file for the proof, so the rows "step 1–5" above follow the outline in its module docstring. For every $Q \in \mathrm{DS}(V,W)$ with T the mass of Q on the 20 core atoms, $d(x) = \deg_G(v) - \deg_H(w)$ and $L = \sum_x q_x |d(x)|$: $4C_{\square}(Q) \ge L + B(s,t) \ge 9(10-T) + \frac{16}{5}T - 16 \ge 16$ for suitable test functions s,t.

2. The finite checks are done by `decide` (no `native_decide`, so no extra axioms). $|x^{\top} Z y| \le 8$ is reduced to $\sum_j |(Zx)_j| \le 8$ over the 3^5 vectors x. The lower bound ≥ 8 is checked for each of the 32 sign patterns r with an explicit table of maximisers `wit`. This replaces the paper's case analysis on $|K|$ up to symmetry.

3. Both graphs live on the same vertex type `Vtx` $= (\mathrm{Fin}\,5 \times \mathrm{Bool}) \oplus \Sigma_i \,\mathrm{Fin}(9(i+1))$; the natural bijection id of the paper is `Equiv.refl`.